Matgeo-4.8.15

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Question

A line l passes through point $(-1, 3, -2)$ and is perpendicular to both the lines $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z+1}{3}$ and $\frac{x+2}{-3} = \frac{y}{2} = \frac{z}{5}$. Find the vector equation of the line l and its distance from the origin.

Solution

Let the direction vector of the required line be $\left(d\right) = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$.

Since the line is perpendicular to the two given lines, the direction vector satisfies

$$\left(d\right)_1^\top \left(d\right) = 0 \quad \Rightarrow \quad 1 \cdot a + 2 \cdot b + 3 \cdot c = 0, \tag{1}$$

$$\left(d\right)_2^\top \left(d\right) = 0 \quad \Rightarrow \quad -3 \cdot a + 2 \cdot b + 5 \cdot c = 0. \tag{2}$$

Solve the system:

$$8b + 14c = 0 \;\; \Rightarrow \;\; b = -\frac{7}{4}c, \tag{3}$$

$$a + 2b + 3c = 0 \;\; \Rightarrow \;\; a = \frac{1}{2}c. \tag{4}$$

Solution

Choosing $c = 4$ to get integer values:

$$(d) = \begin{pmatrix} 2 \\ -7 \\ 4 \end{pmatrix}. \tag{5}$$

Vector equation of the line through $P = (-1, 3, -2)$:

$$(r) = \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix} + t \begin{pmatrix} 2 \\ -7 \\ 4 \end{pmatrix}, \quad t \in \mathbb{R}. \tag{6}$$

Solution

Distance from origin:

$$t = -\frac{(P) \cdot (d)}{(d) \cdot (d)} = -\frac{(-1)(2) + 3(-7) + (-2)(4)}{2^2 + (-7)^2 + 4^2} = \frac{31}{69}, \tag{7}$$

$$(Q) = (P) + t\,(d) = \begin{pmatrix} -\frac{7}{69} \\ \\ -\frac{10}{69} \\ \\ -\frac{14}{69} \end{pmatrix}, \tag{8}$$

$$D = \|\,(Q)\,\| = \sqrt{\frac{345}{69^2}} = \sqrt{\frac{5}{69}}. \tag{9}$$

Conclusion

Conclusion:

The required line is $(r) = \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix} + t \begin{pmatrix} 2 \\ -7 \\ 4 \end{pmatrix}, \quad t \in \mathbb{R}$, and its distance from the origin is $D = \sqrt{\frac{5}{69}}$.

Graphical Representation